Experiment 7

Aim: To study and implement Morphological Analysis of English sentences.

Code:

Morphological Analysis

Install necessary libraries (nltk and spaCy) and download the Brown Corpus.

```
%pip install nltk spacy
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.7) Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.2.1) Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.2) Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2024.11.6)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.1)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12) Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5) Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.13) Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.11) Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.10) Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.6) Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3) Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.1) Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10) Requirement already satisfied: weasel<0.5.0,>=0.1.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.1) Requirement already satisfied: typer<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.17.3) Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2) Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4) Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.11.7) Requirement already satisfied: jinja2 in
```

/usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6) Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.2.0) Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (25.0) Requirement already satisfied: langcodes<4.0.0,>=3.2.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.5.0) Requirement already satisfied: language-data>=1.2 in /usr/local/lib/python3.12/dist-packages (from langcodes<4.0.0,>=3.2.0->spacy) (1.3.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (0.7.0)
Requirement already satisfied: pydantic-core==2.33.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (2.33.2)
Requirement already satisfied: typing-extensions>=4.12.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (4.15.0)
Requirement already satisfied: typing-inspection>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (0.4.1)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.4.3)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (2025.8.3)
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (1.3.0)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (0.1.5)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0.0,>=0.3.0->spacy) (1.5.4)
Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0.0,>=0.3.0->spacy) (13.9.4)
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.1.0->spacy) (0.22.0)

```
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in
/usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.1.0-
>spacy) (7.3.0.post1)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2->spacy) (3.0.2)
Requirement already satisfied: marisa-trie>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from language-data>=1.2-
>langcodes<4.0.0,>=3.2.0->spacy) (1.3.1)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=10.11.0-
>typer<1.0.0,>=0.3.0->spacy) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=10.11.0-
>typer<1.0.0,>=0.3.0->spacy) (2.19.2)
Requirement already satisfied: wrapt in
/usr/local/lib/python3.12/dist-packages (from smart-
open<8.0.0,>=5.2.1->weasel<0.5.0,>=0.1.0->spacy) (1.17.3)
Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0-
>rich>=10.11.0->typer<1.0.0,>=0.3.0->spacy) (0.1.2)
```

```
import nltk nltk.download('brown')
nltk.download('punkt')
nltk.download('averaged_perceptron_tagger')
nltk.download('wordnet') nltk.download('omw-
1.4')
```

```
[nltk_data] Downloading package brown to /root/nltk_data...
[nltk_data]       Unzipping corpora/brown.zip.
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]       Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]         /root/nltk_data...
[nltk_data]       Unzipping taggers/averaged_perceptron_tagger.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
```

```
True
```

```
import spacy
spacy.cli.download("en_core_web_sm")
```

```
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by
```

```
selecting the
'Restart kernel' or 'Restart runtime' option.
```

Data loading and preprocessing

Load the Brown Corpus and perform sentence and word tokenization.

```
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]       Unzipping tokenizers/punkt_tab.zip.
```

```
True
```

```
from nltk.corpus import brown import
nltk

sentences = brown.sents()

tokenized_sentences = []
for sentence in sentences[:10]:          # Take the first 10 sentences as a sample
    tokenized_words = nltk.word_tokenize(" ".join(sentence))
    tokenized_sentences.append(tokenized_words)

print("Sample Tokenized Sentences:")
for i, sentence in enumerate(tokenized_sentences):
    print(f"Sentence {i+1}: {sentence}")
```

```
Sample Tokenized Sentences:
Sentence 1: ['The', 'Fulton', 'County', 'Grand', 'Jury', 'said',
'Friday', 'an', 'investigation', 'of', 'Atlanta', "'s", 'recent',
'primary', 'election', 'produced', '``', 'no', 'evidence', '``',
'that', 'any', 'irregularities', 'took', 'place', '.']
Sentence 2: ['The', 'jury', 'further', 'said', 'in', 'term-end',
'presentments', 'that', 'the', 'City', 'Executive', 'Committee', ',',
'which', 'had', 'over-all', 'charge', 'of', 'the', 'election', ',',
'``', 'deserves', 'the', 'praise', 'and', 'thanks', 'of', 'the',
'City', 'of', 'Atlanta', "''", 'for', 'the', 'manner', 'in', 'which', 'the',
'election', 'was', 'conducted', '.']
Sentence 3: ['The', 'September-October', 'term', 'jury', 'had',
'been', 'charged', 'by', 'Fulton', 'Superior', 'Court', 'Judge',
'Durwood', 'Pye', 'to', 'investigate', 'reports', 'of', 'possible',
'``', 'irregularities', "''", 'in', 'the', 'hard-fought', 'primary',
'which', 'was', 'won', 'by', 'Mayor-nominate', 'Ivan', 'Allen', 'Jr.', '.']
Sentence 4: ['``', 'Only', 'a', 'relative', 'handful', 'of', 'such',
'reports', 'was', 'received', "''", ',', 'the', 'jury', 'said', ',',
"''", 'considering', 'the', 'widespread', 'interest', 'in', 'the',
'election', ',', 'the', 'number', 'of', 'voters', 'and', 'the',
```

```
'size', 'of', 'this', 'city', '``', '.']
Sentence 5: ['The', 'jury', 'said', 'it', 'did', 'find', 'that',
'many', 'of', 'Georgia', "'s", 'registration', 'and', 'election',
'laws', '``', 'are', 'outmoded', 'or', 'inadequate', 'and', 'often',
'ambiguous', '``', '.']
Sentence 6: ['It', 'recommended', 'that', 'Fulton', 'legislators', 'act',
'``', 'to', 'have', 'these', 'laws', 'studied', 'and',
'revised', 'to', 'the', 'end', 'of', 'modernizing', 'and', 'improving',
'them', '``', '.']
Sentence 7: ['The', 'grand', 'jury', 'commented', 'on', 'a', 'number',
'of', 'other', 'topics', ',', 'among', 'them', 'the', 'Atlanta',
'and', 'Fulton', 'County', 'purchasing', 'departments', 'which', 'it',
'said', '``', 'are', 'well', 'operated', 'and', 'follow', 'generally',
'accepted', 'practices', 'which', 'inure', 'to', 'the', 'best',
'interest', 'of', 'both', 'governments', '``', '.'] Sentence
8: ['Merger', 'proposed']
Sentence 9: ['However', ',', 'the', 'jury', 'said', 'it', 'believes',
'``', 'these', 'two', 'offices', 'should', 'be', 'combined', 'to',
'achieve', 'greater', 'efficiency', 'and', 'reduce', 'the', 'cost', 'of',
'administration', '``', '.']
Sentence 10: ['The', 'City', 'Purchasing', 'Department', ',', 'the',
'jury', 'said', ',', '``', 'is', 'lacking', 'in', 'experienced',
'clerical', 'personnel', 'as', 'a', 'result', 'of', 'city', 'personnel',
'policies', '``', '.']
```

Pos tagging

Perform Part-of-speech (POS) tagging on the tokenized words.

```
import nltk
nltk.download('averaged_perceptron_tagger_eng')

from nltk import pos_tag

pos_tagged_sentences = []
for sentence in tokenized_sentences:
    pos_tags = pos_tag(sentence)
    pos_tagged_sentences.append(pos_tags)

print("\nSample POS-Tagged Sentences:")
for i, sentence_tags in enumerate(pos_tagged_sentences[:3]):
    print(f"Sentence {i+1}: {sentence_tags}")
```

```
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger_eng.zip.


Sample POS-Tagged Sentences:
```

```
Sentence 1: [('The', 'DT'), ('Fulton', 'NNP'), ('County', 'NNP'),
('Grand', 'NNP'), ('Jury', 'NNP'), ('said', 'VBD'), ('Friday', 'NNP'),
('an', 'DT'), ('investigation', 'NN'), ('of', 'IN'), ('Atlanta',
'NNP'), ("'s", 'POS'), ('recent', 'JJ'), ('primary', 'JJ'),
('election', 'NN'), ('produced', 'VBD'), ('``', '``'), ('no', 'DT'),
('evidence', 'NN'), ('``', '``'), ('that', 'IN'), ('any', 'DT'),
('irregularities', 'NNS'), ('took', 'VBD'), ('place', 'NN'), ('.', '.')]
Sentence 2: [('The', 'DT'), ('jury', 'NN'), ('further', 'RB'),
('said', 'VBD'), ('in', 'IN'), ('term-end', 'JJ'), ('presentments',
'NNS'), ('that', 'IN'), ('the', 'DT'), ('City', 'NNP'), ('Executive',
'NNP'), ('Committee', 'NNP'), (',', ','), ('which', 'WDT'), ('had',
'VBD'), ('over-all', 'JJ'), ('charge', 'NN'), ('of', 'IN'), ('the',
'DT'), ('election', 'NN'), (',', ','), ('``', '``'), ('deserves',
'VBZ'), ('the', 'DT'), ('praise', 'NN'), ('and', 'CC'), ('thanks',
'NNS'), ('of', 'IN'), ('the', 'DT'), ('City', 'NNP'), ('of', 'IN'),
('Atlanta', 'NNP'), ('``', '``'), ('for', 'IN'), ('the', 'DT'),
('manner', 'NN'), ('in', 'IN'), ('which', 'WDT'), ('the', 'DT'),
('election', 'NN'), ('was', 'VBD'), ('conducted', 'VBN'), ('.', '.')]
Sentence 3: [('The', 'DT'), ('September-October', 'NNP'), ('term',
'NN'), ('jury', 'NN'), ('had', 'VBD'), ('been', 'VBN'), ('charged',
'VBN'), ('by', 'IN'), ('Fulton', 'NNP'), ('Superior', 'NNP'),
('Court', 'NNP'), ('Judge', 'NNP'), ('Durwood', 'NNP'), ('Pye',
'NNP'), ('to', 'TO'), ('investigate', 'VB'), ('reports', 'NNS'),
('of', 'IN'), ('possible', 'JJ'), ('``', '``'), ('irregularities',
'NNS'), ('``', '``'), ('in', 'IN'), ('the', 'DT'), ('hard-fought',
'JJ'), ('primary', 'NN'), ('which', 'WDT'), ('was', 'VBD'), ('won',
'VBN'), ('by', 'IN'), ('Mayor-nominate', 'NNP'), ('Ivan', 'NNP'),
('Allen', 'NNP'), ('Jr.', 'NNP'), ('.', '.')]
```

Lemmatization and stemming

Apply lemmatization and stemming to the words.

```
from nltk.stem import WordNetLemmatizer, PorterStemmer

lemmatizer = WordNetLemmatizer()
stemmer = PorterStemmer()

morphological_analysis_results = []

for sentence in tokenized_sentences:
    sentence_results = []
    for word in sentence:
        lemmatized_word = lemmatizer.lemmatize(word)
        stemmed_word = stemmer.stem(word)
        sentence_results.append({
            "original_word": word,
            "lemmatized_word": lemmatized_word,
```

```
                "stemmed_word": stemmed_word
            })
        morphological_analysis_results.append(sentence_results)

# Print results for a few sample sentences print("\nSample
Morphological Analysis Results:") for i,
sentence_results in
enumerate(morphological_analysis_results[:3]):
        print(f"Sentence {i+1}:")
        for word_results in sentence_results[:5]: # Print only first 5 words for brevity
            print(f"    Original: {word_results['original_word']},
Lemmatized: {word_results['lemmatized_word']}, Stemmed:
{word_results['stemmed_word']}")
```

```
Sample Morphological Analysis Results:
Sentence 1:
  Original: The, Lemmatized: The, Stemmed: the
  Original: Fulton, Lemmatized: Fulton, Stemmed: fulton
  Original: County, Lemmatized: County, Stemmed: counti
  Original: Grand, Lemmatized: Grand, Stemmed: grand
  Original: Jury, Lemmatized: Jury, Stemmed: juri
Sentence 2:
  Original: The, Lemmatized: The, Stemmed: the
  Original: jury, Lemmatized: jury, Stemmed: juri
  Original: further, Lemmatized: further, Stemmed: further
  Original: said, Lemmatized: said, Stemmed: said
  Original: in, Lemmatized: in, Stemmed: in
Sentence 3:
  Original: The, Lemmatized: The, Stemmed: the
  Original: September-October, Lemmatized: September-October, Stemmed:
september-octob
  Original: term, Lemmatized: term, Stemmed: term
  Original: jury, Lemmatized: jury, Stemmed: juri
  Original: had, Lemmatized: had, Stemmed: had
```

Morphological parsing

Use nltk and spaCy to perform morphological parsing.

```
import spacy

# Select a few sample sentences for parsing
sample_sentences_nltk = tokenized_sentences[:3]
sample_sentences_raw_text = [" ".join(sent) for sent in
sample_sentences_nltk]

# NLTK morphological parsing (basic)
```

```
nltk_morph_results = []
for sentence in sample_sentences_nltk:
    sentence_results = []
    for word in sentence:
        # NLTK's direct morphological parsing is limited.
        # We'll get the base form using lemmatization as a proxy for morphological info.
        # For more advanced info, one might need to use WordNet or other resources,
        # but a direct morphological parse like spaCy is not a single NLTK function.
        lemmatized_word = lemmatizer.lemmatize(word)
        sentence_results.append({ "original_word":
            word,
            "nltk_morph_info": f"Base form (lemmatized):
{lemmatized_word}"
        })
    nltk_morph_results.append(sentence_results)

# Load spaCy model
nlp = spacy.load("en_core_web_sm")

# spaCy morphological parsing
spacy_morph_results = []
for sentence_text in sample_sentences_raw_text: doc
    = nlp(sentence_text)
    sentence_results = [] for
    token in doc:
        sentence_results.append({
            "original_word": token.text,
            "spacy_morph_info": str(token.morph) # Morphological features as a string
        })
    spacy_morph_results.append(sentence_results)

# Print results
print("NLTK Morphological Analysis (Basic - using Lemmatization):") for
i, sentence_results in enumerate(nltk_morph_results):
    print(f"Sentence {i+1}:")
    for word_results in sentence_results:
        print(f"    Original: {word_results['original_word']}, Info:
{word_results['nltk_morph_info']}")

print("\nspaCy Morphological Parsing:")
for i, sentence_results in enumerate(spacy_morph_results): print(f"Sentence
    {i+1}:")
    for word_results in sentence_results:
        print(f"    Original: {word_results['original_word']}, Morph:
{word_results['spacy_morph_info']}")
```

NLTK Morphological Analysis (Basic - using Lemmatization):
Sentence 1:
  Original: The, Info: Base form (lemmatized): The
  Original: Fulton, Info: Base form (lemmatized): Fulton
  Original: County, Info: Base form (lemmatized): County
  Original: Grand, Info: Base form (lemmatized): Grand
  Original: Jury, Info: Base form (lemmatized): Jury
  Original: said, Info: Base form (lemmatized): said
  Original: Friday, Info: Base form (lemmatized): Friday
  Original: an, Info: Base form (lemmatized): an
  Original: investigation, Info: Base form (lemmatized): investigation
  Original: of, Info: Base form (lemmatized): of
  Original: Atlanta, Info: Base form (lemmatized): Atlanta
  Original: 's, Info: Base form (lemmatized): 's
  Original: recent, Info: Base form (lemmatized): recent
  Original: primary, Info: Base form (lemmatized): primary
  Original: election, Info: Base form (lemmatized): election
  Original: produced, Info: Base form (lemmatized): produced
  Original: ``, Info: Base form (lemmatized): ``
  Original: no, Info: Base form (lemmatized): no
  Original: evidence, Info: Base form (lemmatized): evidence
  Original: ``, Info: Base form (lemmatized): ``
  Original: that, Info: Base form (lemmatized): that
  Original: any, Info: Base form (lemmatized): any
  Original: irregularities, Info: Base form (lemmatized): irregularity
  Original: took, Info: Base form (lemmatized): took
  Original: place, Info: Base form (lemmatized): place
  Original: ., Info: Base form (lemmatized): .
Sentence 2:
  Original: The, Info: Base form (lemmatized): The
  Original: jury, Info: Base form (lemmatized): jury
  Original: further, Info: Base form (lemmatized): further
  Original: said, Info: Base form (lemmatized): said
  Original: in, Info: Base form (lemmatized): in
  Original: term-end, Info: Base form (lemmatized): term-end
  Original: presentments, Info: Base form (lemmatized): presentment
  Original: that, Info: Base form (lemmatized): that
  Original: the, Info: Base form (lemmatized): the
  Original: City, Info: Base form (lemmatized): City
  Original: Executive, Info: Base form (lemmatized): Executive
  Original: Committee, Info: Base form (lemmatized): Committee
  Original: ,, Info: Base form (lemmatized): ,
  Original: which, Info: Base form (lemmatized): which
  Original: had, Info: Base form (lemmatized): had
  Original: over-all, Info: Base form (lemmatized): over-all
  Original: charge, Info: Base form (lemmatized): charge
  Original: of, Info: Base form (lemmatized): of
  Original: the, Info: Base form (lemmatized): the
  Original: election, Info: Base form (lemmatized): election
  Original: '', Info: Base form (lemmatized): '

Original: ``, Info: Base form (lemmatized): ``
Original: deserves, Info: Base form (lemmatized): deserves
Original: the, Info: Base form (lemmatized): the
Original: praise, Info: Base form (lemmatized): praise
Original: and, Info: Base form (lemmatized): and
Original: thanks, Info: Base form (lemmatized): thanks
Original: of, Info: Base form (lemmatized): of Original: the, Info: Base form (lemmatized): the Original: City, Info: Base form (lemmatized): City Original: of, Info: Base form (lemmatized): of
Original: Atlanta, Info: Base form (lemmatized): Atlanta
Original: ``, Info: Base form (lemmatized): ``
Original: for, Info: Base form (lemmatized): for Original: the, Info: Base form (lemmatized): the Original: manner, Info: Base form (lemmatized): manner Original: in, Info: Base form (lemmatized): in Original: which, Info: Base form (lemmatized): which Original: the, Info: Base form (lemmatized): the
Original: election, Info: Base form (lemmatized): election
Original: was, Info: Base form (lemmatized): wa
Original: conducted, Info: Base form (lemmatized): conducted Original: .., Info: Base form (lemmatized): .

Sentence 3:
Original: The, Info: Base form (lemmatized): The
Original: September-October, Info: Base form (lemmatized): September-October
Original: term, Info: Base form (lemmatized): term Original: jury, Info: Base form (lemmatized): jury Original: had, Info: Base form (lemmatized): had Original: been, Info: Base form (lemmatized): been Original: charged, Info: Base form (lemmatized): charged Original: by, Info: Base form (lemmatized): by
Original: Fulton, Info: Base form (lemmatized): Fulton
Original: Superior, Info: Base form (lemmatized): Superior
Original: Court, Info: Base form (lemmatized): Court Original: Judge, Info: Base form (lemmatized): Judge Original: Durwood, Info: Base form (lemmatized): Durwood Original: Pye, Info: Base form (lemmatized): Pye
Original: to, Info: Base form (lemmatized): to
Original: investigate, Info: Base form (lemmatized): investigate
Original: reports, Info: Base form (lemmatized): report Original: of, Info: Base form (lemmatized): of
Original: possible, Info: Base form (lemmatized): possible
Original: ``, Info: Base form (lemmatized): ``
Original: irregularities, Info: Base form (lemmatized): irregularity
Original: ``, Info: Base form (lemmatized): ``
Original: in, Info: Base form (lemmatized): in
Original: the, Info: Base form (lemmatized): the
Original: hard-fought, Info: Base form (lemmatized): hard-fought Original: primary, Info: Base form (lemmatized): primary

Original: which, Info: Base form (lemmatized): which
Original: was, Info: Base form (lemmatized): wa
Original: won, Info: Base form (lemmatized): won
Original: by, Info: Base form (lemmatized): by
Original: Mayor-nominate, Info: Base form (lemmatized): Mayor-nominate
Original: Ivan, Info: Base form (lemmatized): Ivan
Original: Allen, Info: Base form (lemmatized): Allen
Original: Jr., Info: Base form (lemmatized): Jr.
Original: ., Info: Base form (lemmatized): .

spaCy Morphological Parsing:
Sentence 1:
Original: The, Morph: Definite=Def|PronType=Art
Original: Fulton, Morph: Number=Sing
Original: County, Morph: Number=Sing
Original: Grand, Morph: Number=Sing
Original: Jury, Morph: Number=Sing
Original: said, Morph: Tense=Past|VerbForm=Fin
Original: Friday, Morph: Number=Sing
Original: an, Morph: Definite=Ind|PronType=Art
Original: investigation, Morph: Number=Sing
Original: of, Morph:
Original: Atlanta, Morph: Number=Sing
Original: 's, Morph:
Original: recent, Morph: Degree=Pos
Original: primary, Morph: Degree=Pos
Original: election, Morph: Number=Sing
Original: produced, Morph: Aspect=Perf|Tense=Past|VerbForm=Part
Original: `, Morph: PunctSide=Ini|PunctType=Quot
Original: `, Morph: PunctSide=Ini|PunctType=Quot
Original: no, Morph:
Original: evidence, Morph: Number=Sing
Original: `, Morph: PunctSide=Ini|PunctType=Quot
Original: `, Morph: PunctSide=Ini|PunctType=Quot
Original: that, Morph:
Original: any, Morph:
Original: irregularities, Morph: Number=Plur
Original: took, Morph: Tense=Past|VerbForm=Fin
Original: place, Morph: Number=Sing
Original: ., Morph: PunctType=Peri
Sentence 2:
Original: The, Morph: Definite=Def|PronType=Art
Original: jury, Morph: Number=Sing
Original: further, Morph:
Original: said, Morph: Tense=Past|VerbForm=Fin
Original: in, Morph:
Original: term, Morph: Number=Sing
Original: -, Morph: PunctType=Dash
Original: end, Morph: Number=Sing

Original: presentments, Morph: Number=Plur Original:
that, Morph:
Original: the, Morph: Definite=Def|PronType=Art
Original: City, Morph: Number=Sing
Original: Executive, Morph: Number=Sing
Original: Committee, Morph: Number=Sing
Original: ,, Morph: PunctType=Comm
Original: which, Morph:
Original: had, Morph: Tense=Past|VerbForm=Fin
Original: over, Morph:
Original: -, Morph: PunctType=Dash
Original: all, Morph:
Original: charge, Morph: Number=Sing
Original: of, Morph:
Original: the, Morph: Definite=Def|PronType=Art
Original: election, Morph: Number=Sing Original:
,, Morph: PunctType=Comm
Original: `, Morph: PunctSide=Ini|PunctType=Quot Original:
`, Morph: PunctSide=Ini|PunctType=Quot
Original: deserves, Morph: Number=Sing|Person=3|Tense=Pres| VerbForm=Fin
Original: the, Morph: Definite=Def|PronType=Art
Original: praise, Morph: Number=Sing
Original: and, Morph: ConjType=Cmp Original:
thanks, Morph: Number=Plur Original: of,
Morph:
Original: the, Morph: Definite=Def|PronType=Art
Original: City, Morph: Number=Sing
Original: of, Morph:
Original: Atlanta, Morph: Number=Sing
Original: `, Morph: PunctSide=Ini|PunctType=Quot
Original: `, Morph: PunctSide=Ini|PunctType=Quot
Original: for, Morph:
Original: the, Morph: Definite=Def|PronType=Art
Original: manner, Morph: Number=Sing
Original: in, Morph:
Original: which, Morph:
Original: the, Morph: Definite=Def|PronType=Art
Original: election, Morph: Number=Sing
Original: was, Morph: Mood=Ind|Number=Sing|Person=3|Tense=Past| VerbForm=Fin
Original: conducted, Morph: Aspect=Perf|Tense=Past|VerbForm=Part Original: .,
Morph: PunctType=Peri
Sentence 3:
Original: The, Morph: Definite=Def|PronType=Art
Original: September, Morph: Number=Sing Original: -,
Morph: PunctType=Dash
Original: October, Morph: Number=Sing
Original: term, Morph: Number=Sing
Original: jury, Morph: Number=Sing

```
  Original: had, Morph: Tense=Past|VerbForm=Fin
  Original: been, Morph: Tense=Past|VerbForm=Part
  Original: charged, Morph: Aspect=Perf|Tense=Past|VerbForm=Part
  Original: by, Morph:
  Original: Fulton, Morph: Number=Sing
  Original: Superior, Morph: Number=Sing
  Original: Court, Morph: Number=Sing
  Original: Judge, Morph: Number=Sing
  Original: Durwood, Morph: Number=Sing
  Original: Pye, Morph: Number=Sing
  Original: to, Morph:
  Original: investigate, Morph: VerbForm=Inf
  Original: reports, Morph: Number=Plur
  Original: of, Morph:
  Original: possible, Morph: Degree=Pos
  Original: `, Morph: PunctSide=Ini|PunctType=Quot
  Original: `, Morph: PunctSide=Ini|PunctType=Quot
  Original: irregularities, Morph: Number=Plur
  Original: `, Morph: PunctSide=Ini|PunctType=Quot
  Original: `, Morph: PunctSide=Ini|PunctType=Quot
  Original: in, Morph:
  Original: the, Morph: Definite=Def|PronType=Art
  Original: hard, Morph:
  Original: -, Morph: PunctType=Dash
  Original: fought, Morph: Aspect=Perf|Tense=Past|VerbForm=Part
  Original: primary, Morph: Number=Sing
  Original: which, Morph:
  Original: was, Morph: Mood=Ind|Number=Sing|Person=3|Tense=Past|
VerbForm=Fin
  Original: won, Morph: Aspect=Perf|Tense=Past|VerbForm=Part
  Original: by, Morph:
  Original: Mayor, Morph: Number=Sing
  Original: -, Morph: PunctType=Dash
  Original: nominate, Morph: Number=Sing
  Original: Ivan, Morph: Number=Sing
  Original: Allen, Morph: Number=Sing
  Original: Jr., Morph: Number=Sing
  Original: ., Morph: PunctType=Peri
```

Visualization

Visualize sample sentences showing their morphological structure.

```
sample_sentence_indices = [0, 1, 2] # Select the first three sentences
as samples

print("Visualizing Morphological Structure:")

for i in sample_sentence_indices:
```

```
    print(f"\n--- Sentence {i+1} ---")

    print("NLTK Morphological Structure (using Lemmatization):") if
    i < len(nltk_morph_results):
        for word_results in nltk_morph_results[i]:
            print(f"    Original: {word_results['original_word']}, Info:
{word_results['nltk_morph_info']}")
    else:
        print("    NLTK results not available for this sentence index.")


    print("\nspaCy Morphological Structure:") if
    i < len(spacy_morph_results):
        for word_results in spacy_morph_results[i]:
            print(f"    Original: {word_results['original_word']},
Morph: {word_results['spacy_morph_info']}")
    else:
        print("    spaCy results not available for this sentence
index.")
```

```
Visualizing Morphological Structure:

--- Sentence 1 ---
NLTK Morphological Structure (using Lemmatization):
  Original: The, Info: Base form (lemmatized): The
  Original: Fulton, Info: Base form (lemmatized): Fulton
  Original: County, Info: Base form (lemmatized): County
  Original: Grand, Info: Base form (lemmatized): Grand
  Original: Jury, Info: Base form (lemmatized): Jury
  Original: said, Info: Base form (lemmatized): said
  Original: Friday, Info: Base form (lemmatized): Friday
  Original: an, Info: Base form (lemmatized): an
  Original: investigation, Info: Base form (lemmatized): investigation
  Original: of, Info: Base form (lemmatized): of
  Original: Atlanta, Info: Base form (lemmatized): Atlanta
  Original: 's, Info: Base form (lemmatized): 's
  Original: recent, Info: Base form (lemmatized): recent
  Original: primary, Info: Base form (lemmatized): primary
  Original: election, Info: Base form (lemmatized): election
  Original: produced, Info: Base form (lemmatized): produced
  Original: ``, Info: Base form (lemmatized): ``
  Original: no, Info: Base form (lemmatized): no
  Original: evidence, Info: Base form (lemmatized): evidence
  Original: ``, Info: Base form (lemmatized): ``
  Original: that, Info: Base form (lemmatized): that
  Original: any, Info: Base form (lemmatized): any
  Original: irregularities, Info: Base form (lemmatized): irregularity
  Original: took, Info: Base form (lemmatized): took
  Original: place, Info: Base form (lemmatized): place
  Original: ., Info: Base form (lemmatized): .
```

```
spaCy Morphological Structure:
  Original: The, Morph: Definite=Def|PronType=Art
  Original: Fulton, Morph: Number=Sing
  Original: County, Morph: Number=Sing
  Original: Grand, Morph: Number=Sing
  Original: Jury, Morph: Number=Sing
  Original: said, Morph: Tense=Past|VerbForm=Fin
  Original: Friday, Morph: Number=Sing
  Original: an, Morph: Definite=Ind|PronType=Art
  Original: investigation, Morph: Number=Sing
  Original: of, Morph:
  Original: Atlanta, Morph: Number=Sing
  Original: 's, Morph:
  Original: recent, Morph: Degree=Pos
  Original: primary, Morph: Degree=Pos
  Original: election, Morph: Number=Sing
  Original: produced, Morph: Aspect=Perf|Tense=Past|VerbForm=Part
  Original: `, Morph: PunctSide=Ini|PunctType=Quot
  Original: `, Morph: PunctSide=Ini|PunctType=Quot
  Original: no, Morph:
  Original: evidence, Morph: Number=Sing
  Original: `, Morph: PunctSide=Ini|PunctType=Quot
  Original: `, Morph: PunctSide=Ini|PunctType=Quot
  Original: that, Morph:
  Original: any, Morph:
  Original: irregularities, Morph: Number=Plur
  Original: took, Morph: Tense=Past|VerbForm=Fin
  Original: place, Morph: Number=Sing
  Original: ., Morph: PunctType=Peri

--- Sentence 2 ---
NLTK Morphological Structure (using Lemmatization):
  Original: The, Info: Base form (lemmatized): The
  Original: jury, Info: Base form (lemmatized): jury
  Original: further, Info: Base form (lemmatized): further
  Original: said, Info: Base form (lemmatized): said
  Original: in, Info: Base form (lemmatized): in
  Original: term-end, Info: Base form (lemmatized): term-end
  Original: presentments, Info: Base form (lemmatized): presentment
  Original: that, Info: Base form (lemmatized): that
  Original: the, Info: Base form (lemmatized): the
  Original: City, Info: Base form (lemmatized): City
  Original: Executive, Info: Base form (lemmatized): Executive
  Original: Committee, Info: Base form (lemmatized): Committee
  Original: ,, Info: Base form (lemmatized): ,
  Original: which, Info: Base form (lemmatized): which
  Original: had, Info: Base form (lemmatized): had
  Original: over-all, Info: Base form (lemmatized): over-all
```

Original: charge, Info: Base form (lemmatized): charge
Original: of, Info: Base form (lemmatized): of
Original: the, Info: Base form (lemmatized): the
Original: election, Info: Base form (lemmatized): election
Original: ,, Info: Base form (lemmatized): ,
Original: ``, Info: Base form (lemmatized): ``
Original: deserves, Info: Base form (lemmatized): deserves
Original: the, Info: Base form (lemmatized): the
Original: praise, Info: Base form (lemmatized): praise
Original: and, Info: Base form (lemmatized): and
Original: thanks, Info: Base form (lemmatized): thanks
Original: of, Info: Base form (lemmatized): of
Original: the, Info: Base form (lemmatized): the
Original: City, Info: Base form (lemmatized): City
Original: of, Info: Base form (lemmatized): of
Original: Atlanta, Info: Base form (lemmatized): Atlanta
Original: ``, Info: Base form (lemmatized): ``
Original: for, Info: Base form (lemmatized): for
Original: the, Info: Base form (lemmatized): the
Original: manner, Info: Base form (lemmatized): manner
Original: in, Info: Base form (lemmatized): in
Original: which, Info: Base form (lemmatized): which
Original: the, Info: Base form (lemmatized): the
Original: election, Info: Base form (lemmatized): election
Original: was, Info: Base form (lemmatized): wa
Original: conducted, Info: Base form (lemmatized): conducted
Original: ., Info: Base form (lemmatized): .

spaCy Morphological Structure:
Original: The, Morph: Definite=Def|PronType=Art
Original: jury, Morph: Number=Sing
Original: further, Morph:
Original: said, Morph: Tense=Past|VerbForm=Fin
Original: in, Morph:
Original: term, Morph: Number=Sing
Original: -, Morph: PunctType=Dash
Original: end, Morph: Number=Sing
Original: presentments, Morph: Number=Plur
Original: that, Morph:
Original: the, Morph: Definite=Def|PronType=Art
Original: City, Morph: Number=Sing
Original: Executive, Morph: Number=Sing
Original: Committee, Morph: Number=Sing
Original: ,, Morph: PunctType=Comm
Original: which, Morph:
Original: had, Morph: Tense=Past|VerbForm=Fin
Original: over, Morph:
Original: -, Morph: PunctType=Dash
Original: all, Morph:

Original: charge, Morph: Number=Sing
Original: of, Morph:
Original: the, Morph: Definite=Def|PronType=Art
Original: election, Morph: Number=Sing
Original: ,, Morph: PunctType=Comm
Original: `, Morph: PunctSide=Ini|PunctType=Quot
Original: `, Morph: PunctSide=Ini|PunctType=Quot
Original: deserves, Morph: Number=Sing|Person=3|Tense=Pres|VerbForm=Fin
Original: the, Morph: Definite=Def|PronType=Art
Original: praise, Morph: Number=Sing
Original: and, Morph: ConjType=Cmp
Original: thanks, Morph: Number=Plur
Original: of, Morph:
Original: the, Morph: Definite=Def|PronType=Art
Original: City, Morph: Number=Sing
Original: of, Morph:
Original: Atlanta, Morph: Number=Sing
Original: `, Morph: PunctSide=Ini|PunctType=Quot
Original: `, Morph: PunctSide=Ini|PunctType=Quot
Original: for, Morph:
Original: the, Morph: Definite=Def|PronType=Art
Original: manner, Morph: Number=Sing
Original: in, Morph:
Original: which, Morph:
Original: the, Morph: Definite=Def|PronType=Art
Original: election, Morph: Number=Sing
Original: was, Morph: Mood=Ind|Number=Sing|Person=3|Tense=Past|VerbForm=Fin
Original: conducted, Morph: Aspect=Perf|Tense=Past|VerbForm=Part
Original: ., Morph: PunctType=Peri

--- Sentence 3 ---
NLTK Morphological Structure (using Lemmatization):
Original: The, Info: Base form (lemmatized): The
Original: September-October, Info: Base form (lemmatized): September-October
Original: term, Info: Base form (lemmatized): term
Original: jury, Info: Base form (lemmatized): jury
Original: had, Info: Base form (lemmatized): had
Original: been, Info: Base form (lemmatized): been
Original: charged, Info: Base form (lemmatized): charged
Original: by, Info: Base form (lemmatized): by
Original: Fulton, Info: Base form (lemmatized): Fulton
Original: Superior, Info: Base form (lemmatized): Superior
Original: Court, Info: Base form (lemmatized): Court
Original: Judge, Info: Base form (lemmatized): Judge
Original: Durwood, Info: Base form (lemmatized): Durwood
Original: Pye, Info: Base form (lemmatized): Pye

Original: to, Info: Base form (lemmatized): to
Original: investigate, Info: Base form (lemmatized): investigate
Original: reports, Info: Base form (lemmatized): report
Original: of, Info: Base form (lemmatized): of
Original: possible, Info: Base form (lemmatized): possible
Original: ``, Info: Base form (lemmatized): ``
Original: irregularities, Info: Base form (lemmatized): irregularity
Original: ``, Info: Base form (lemmatized): ``
Original: in, Info: Base form (lemmatized): in
Original: the, Info: Base form (lemmatized): the
Original: hard-fought, Info: Base form (lemmatized): hard-fought
Original: primary, Info: Base form (lemmatized): primary
Original: which, Info: Base form (lemmatized): which
Original: was, Info: Base form (lemmatized): wa
Original: won, Info: Base form (lemmatized): won
Original: by, Info: Base form (lemmatized): by
Original: Mayor-nominate, Info: Base form (lemmatized): Mayor-nominate
Original: Ivan, Info: Base form (lemmatized): Ivan
Original: Allen, Info: Base form (lemmatized): Allen
Original: Jr., Info: Base form (lemmatized): Jr.
Original: ., Info: Base form (lemmatized): .

spaCy Morphological Structure:
Original: The, Morph: Definite=Def|PronType=Art
Original: September, Morph: Number=Sing
Original: -, Morph: PunctType=Dash
Original: October, Morph: Number=Sing
Original: term, Morph: Number=Sing
Original: jury, Morph: Number=Sing
Original: had, Morph: Tense=Past|VerbForm=Fin
Original: been, Morph: Tense=Past|VerbForm=Part
Original: charged, Morph: Aspect=Perf|Tense=Past|VerbForm=Part
Original: by, Morph:
Original: Fulton, Morph: Number=Sing
Original: Superior, Morph: Number=Sing
Original: Court, Morph: Number=Sing
Original: Judge, Morph: Number=Sing
Original: Durwood, Morph: Number=Sing
Original: Pye, Morph: Number=Sing
Original: to, Morph:
Original: investigate, Morph: VerbForm=Inf
Original: reports, Morph: Number=Plur
Original: of, Morph:
Original: possible, Morph: Degree=Pos
Original: `, Morph: PunctSide=Ini|PunctType=Quot
Original: `, Morph: PunctSide=Ini|PunctType=Quot
Original: irregularities, Morph: Number=Plur
Original: `, Morph: PunctSide=Ini|PunctType=Quot

```
  Original: `, Morph: PunctSide=Ini|PunctType=Quot
  Original: in, Morph:
  Original: the, Morph: Definite=Def|PronType=Art
  Original: hard, Morph:
  Original: -, Morph: PunctType=Dash
  Original: fought, Morph: Aspect=Perf|Tense=Past|VerbForm=Part
  Original: primary, Morph: Number=Sing
  Original: which, Morph:
  Original: was, Morph: Mood=Ind|Number=Sing|Person=3|Tense=Past|
VerbForm=Fin
  Original: won, Morph: Aspect=Perf|Tense=Past|VerbForm=Part
  Original: by, Morph:
  Original: Mayor, Morph: Number=Sing
  Original: -, Morph: PunctType=Dash
  Original: nominate, Morph: Number=Sing
  Original: Ivan, Morph: Number=Sing
  Original: Allen, Morph: Number=Sing
  Original: Jr., Morph: Number=Sing
  Original: ., Morph: PunctType=Peri
```

```
for i in sample_sentence_indices:
    if i < len(nltk_morph_results) and i < len(spacy_morph_results): if i
        == 2: #Continue printing for the third sentence
            print(f"\n--- Sentence {i+1} ---")

            print("NLTK Morphological Structure (using
Lemmatization):")
            for word_results in nltk_morph_results[i]:
                print(f"    Original: {word_results['original_word']},
Info: {word_results['nltk_morph_info']}")

            print("\nspaCy Morphological Structure:") for
            word_results in spacy_morph_results[i]:
                print(f"    Original: {word_results['original_word']},
Morph: {word_results['spacy_morph_info']}")
```

```
--- Sentence 3 ---
NLTK Morphological Structure (using Lemmatization):
  Original: The, Info: Base form (lemmatized): The
  Original: September-October, Info: Base form (lemmatized):
September-October
  Original: term, Info: Base form (lemmatized): term
  Original: jury, Info: Base form (lemmatized): jury
  Original: had, Info: Base form (lemmatized): had
  Original: been, Info: Base form (lemmatized): been
  Original: charged, Info: Base form (lemmatized): charged
  Original: by, Info: Base form (lemmatized): by
  Original: Fulton, Info: Base form (lemmatized): Fulton
  Original: Superior, Info: Base form (lemmatized): Superior
```

Original: Court, Info: Base form (lemmatized): Court
Original: Judge, Info: Base form (lemmatized): Judge
Original: Durwood, Info: Base form (lemmatized): Durwood
Original: Pye, Info: Base form (lemmatized): Pye
Original: to, Info: Base form (lemmatized): to
Original: investigate, Info: Base form (lemmatized): investigate
Original: reports, Info: Base form (lemmatized): report
Original: of, Info: Base form (lemmatized): of
Original: possible, Info: Base form (lemmatized): possible
Original: ``, Info: Base form (lemmatized): ``
Original: irregularities, Info: Base form (lemmatized): irregularity
Original: ``, Info: Base form (lemmatized): ``
Original: in, Info: Base form (lemmatized): in
Original: the, Info: Base form (lemmatized): the
Original: hard-fought, Info: Base form (lemmatized): hard-fought
Original: primary, Info: Base form (lemmatized): primary
Original: which, Info: Base form (lemmatized): which
Original: was, Info: Base form (lemmatized): wa
Original: won, Info: Base form (lemmatized): won
Original: by, Info: Base form (lemmatized): by
Original: Mayor-nominate, Info: Base form (lemmatized): Mayor-nominate
Original: Ivan, Info: Base form (lemmatized): Ivan
Original: Allen, Info: Base form (lemmatized): Allen
Original: Jr., Info: Base form (lemmatized): Jr.
Original: ., Info: Base form (lemmatized): .

spaCy Morphological Structure:
Original: The, Morph: Definite=Def|PronType=Art
Original: September, Morph: Number=Sing
Original: -, Morph: PunctType=Dash
Original: October, Morph: Number=Sing
Original: term, Morph: Number=Sing
Original: jury, Morph: Number=Sing
Original: had, Morph: Tense=Past|VerbForm=Fin
Original: been, Morph: Tense=Past|VerbForm=Part
Original: charged, Morph: Aspect=Perf|Tense=Past|VerbForm=Part
Original: by, Morph:
Original: Fulton, Morph: Number=Sing
Original: Superior, Morph: Number=Sing
Original: Court, Morph: Number=Sing
Original: Judge, Morph: Number=Sing
Original: Durwood, Morph: Number=Sing
Original: Pye, Morph: Number=Sing
Original: to, Morph:
Original: investigate, Morph: VerbForm=Inf
Original: reports, Morph: Number=Plur
Original: of, Morph:
Original: possible, Morph: Degree=Pos

Original: `, Morph: PunctSide=Ini|PunctType=Quot
Original: `, Morph: PunctSide=Ini|PunctType=Quot
Original: irregularities, Morph: Number=Plur
Original: `, Morph: PunctSide=Ini|PunctType=Quot
Original: `, Morph: PunctSide=Ini|PunctType=Quot
Original: in, Morph:
Original: the, Morph: Definite=Def|PronType=Art
Original: hard, Morph:
Original: -, Morph: PunctType=Dash
Original: fought, Morph: Aspect=Perf|Tense=Past|VerbForm=Part
Original: primary, Morph: Number=Sing
Original: which, Morph:
Original: was, Morph: Mood=Ind|Number=Sing|Person=3|Tense=Past|
VerbForm=Fin
Original: won, Morph: Aspect=Perf|Tense=Past|VerbForm=Part
Original: by, Morph:
Original: Mayor, Morph: Number=Sing
Original: -, Morph: PunctType=Dash
Original: nominate, Morph: Number=Sing
Original: Ivan, Morph: Number=Sing
Original: Allen, Morph: Number=Sing
Original: Jr., Morph: Number=Sing
Original: ., Morph: PunctType=Peri